MSG No. 131.442 $P4'_2/m'mc'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `m'mm`.

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,7,8,11,12,13,14]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,4,5,6,9,10,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `m'mm`.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,5,6,9,10,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `m'mm`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[3,4,5,6,9,10,15,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `m'mm`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,4,5,6,9,10,15,16]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4m2`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,10,11,12,13,14,15,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4m2`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,11,12,13,14,15,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `2mm`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,7,8]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[3,4,5,6]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[9,10,15,16]`
4	$[0,\ 0,\ -z]$	`[11,12,13,14]`

Table 8: Wyckoff site: `4h`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[11,12,13,14]`

Table 9: Wyckoff site: `4i`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[0,\ \frac{1}{2},\ -z\right]$	`[11,12,13,14]`

Table 10: Wyckoff site: `4j`, site symmetry: `m'2'm`.

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,8,11,14]`
2	$[-x,\ 0,\ 0]$	`[2,7,12,13]`
3	$\left[0,\ x,\ \frac{1}{2}\right]$	`[3,6,9,16]`
4	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[4,5,10,15]`

Table 11: Wyckoff site: `4k`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,8,11,14]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,7,12,13]`
3	$\left[\frac{1}{2},\ x,\ 0\right]$	`[3,6,9,16]`
4	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[4,5,10,15]`

Table 12: Wyckoff site: `4l`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,8,11,14]`
2	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[2,7,12,13]`
3	$[0,\ x,\ 0]$	`[3,6,9,16]`

continued ...

Table 12

No.	position	mapping
4	$[0, -x, 0]$	`[4,5,10,15]`

Table 13: Wyckoff site: `4m`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	`[1,8,11,14]`
2	$\left[-x, \frac{1}{2}, 0\right]$	`[2,7,12,13]`
3	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[3,6,9,16]`
4	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[4,5,10,15]`

Table 14: Wyckoff site: `8n`, site symmetry: `..2`

No.	position	mapping
1	$\left[x, x, \frac{1}{4}\right]$	`[1,3]`
2	$\left[-x, -x, \frac{1}{4}\right]$	`[2,4]`
3	$\left[x, -x, \frac{1}{4}\right]$	`[5,8]`
4	$\left[-x, x, \frac{1}{4}\right]$	`[6,7]`
5	$\left[-x, x, \frac{3}{4}\right]$	`[9,12]`
6	$\left[x, -x, \frac{3}{4}\right]$	`[10,11]`
7	$\left[-x, -x, \frac{3}{4}\right]$	`[13,15]`
8	$\left[x, x, \frac{3}{4}\right]$	`[14,16]`

Table 15: Wyckoff site: `8o`, site symmetry: `.m.`

No.	position	mapping
1	$[0, y, z]$	`[1,7]`
2	$[0, -y, z]$	`[2,8]`
3	$\left[y, 0, \frac{1}{2} - z\right]$	`[3,5]`
4	$\left[-y, 0, \frac{1}{2} - z\right]$	`[4,6]`
5	$\left[-y, 0, z + \frac{1}{2}\right]$	`[9,15]`
6	$\left[y, 0, z + \frac{1}{2}\right]$	`[10,16]`
7	$[0, -y, -z]$	`[11,13]`
8	$[0, y, -z]$	`[12,14]`

Table 16: Wyckoff site: `8p`, site symmetry: `.m.`

No.	position	mapping
1	$\left[\frac{1}{2}, y, z\right]$	`[1,7]`
2	$\left[\frac{1}{2}, -y, z\right]$	`[2,8]`
3	$\left[y, \frac{1}{2}, \frac{1}{2} - z\right]$	`[3,5]`
4	$\left[-y, \frac{1}{2}, \frac{1}{2} - z\right]$	`[4,6]`
5	$\left[-y, \frac{1}{2}, z + \frac{1}{2}\right]$	`[9,15]`

continued ...

Table 16

No.	position	mapping
6	$\left[y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[10,16]
7	$\left[\frac{1}{2},\, -y,\, -z\right]$	[11,13]
8	$\left[\frac{1}{2},\, y,\, -z\right]$	[12,14]

Table 17: Wyckoff site: 8q, site symmetry: m'..

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,14]
2	$[-x,\, -y,\, 0]$	[2,13]
3	$\left[y,\, x,\, \frac{1}{2}\right]$	[3,16]
4	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[4,15]
5	$\left[y,\, -x,\, \frac{1}{2}\right]$	[5,10]
6	$\left[-y,\, x,\, \frac{1}{2}\right]$	[6,9]
7	$[-x,\, y,\, 0]$	[7,12]
8	$[x,\, -y,\, 0]$	[8,11]

Table 18: Wyckoff site: 16r, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[3]
4	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[4]
5	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[5]
6	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[6]
7	$[-x,\, y,\, z]$	[7]
8	$[x,\, -y,\, z]$	[8]
9	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[9]
10	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[10]
11	$[x,\, -y,\, -z]$	[11]
12	$[-x,\, y,\, -z]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]